AMC 12A 2023 Problem 17

Strategic Analysis

Using Framework v2.1

Problem Statement

Flora the frog starts at 0 on the number line and makes a sequence of jumps to the right. In any one jump, independent of previous jumps, Flora leaps a positive integer distance m with probability $\frac{1}{2^m}$.

What is the probability that Flora will eventually land at 10?

(A) $\frac{5}{512}$ **(B)** $\frac{45}{1024}$ **(C)** $\frac{127}{1024}$ **(D)** $\frac{511}{1024}$ **(E)** $\frac{1}{2}$

Problem Configuration Analysis

1. Problem Configuration Analysis

A. Problem Classification

- **Competition**: AMC 12A 2023
- **Problem Number**: 17
- **Difficulty Band**: Upper-mid range (Problem 17/25)
- **Expected Time**: 3-5 minutes
- **Point Value**: 6 points (standard AMC 12 scoring)

B. Mathematical Domain Classification

- **Primary Domain**: Probability
- **Secondary Domain**: Combinatorics / Recursion
- **Tertiary Domain**: Series / Generating Functions
- **Cross-Domain Potential**: Very High (probability, counting, algebra, induction)

C. Problem Type

- **Format**: Multiple choice (probability value)

- **Question Type**: "What is the probability..."
- **Core Task**: Calculate probability of landing on specific position
- **Answer Form**: Simplified fraction

D. Given Information

- Starting position: 0 on the number line
- Jump distribution: P(jump distance m) $= \frac{1}{2^m}$ for positive integer m
- Jumps are independent
- Flora only jumps to the right (positive direction)
- Target: Land on position 10

E. Constraints

- Jumps must be positive integers
- Can only move right (no backtracking)
- Once past 10, cannot return
- Must land exactly on 10 (not just pass through)

F. Objective

- **Find**: Probability of eventually landing on 10
- **Output**: Select correct probability from choices

G. Key Structural Features

- **Probability distribution**: Sum of $\frac{1}{2^m}$ for $m \geq 1$ equals 1
- **Recursive structure**: Can land on 10 from any position < 10
- **Counting interpretation**: Count paths that land on 10
- **Symmetry**: Problem has elegant structure suggesting simple answer
- **Answer choices**: (E) $\frac{1}{2}$ is suspiciously simple

H. Strategic Orientation

- **Primary Strategy**: Look for pattern or recursive formula
- **Key Insight**: Probability of landing on any $n \geq 1$ is constant
- **Expected Difficulty**: Moderate - requires insight or systematic calculation
- **Time Pressure**: Should be completed in 3-5 minutes

2. Strategic Framework Application

A. Psychological Strategies

- **Confidence Calibration**:
  - Upper-mid difficulty problem
  - Multiple solution approaches available
  - Simple answer (E) suggests elegant solution
- **Pattern Recognition**:
  - Answer $\frac{1}{2}$ is very clean
  - Probability distribution sums to 1
  - Recursive/self-similar structure
- **Problem Familiarity**:
  - Related to random walk problems
  - Connection to geometric series
  - Stars and bars counting technique

B. Investigation Strategies

- **Initial Exploration**:
  - Calculate P(landing on 1): only one way, jump 1, probability $\frac{1}{2}$
  - Calculate P(landing on 2): from 0→2 directly ($\frac{1}{4}$) or 0→1→2 ($\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$), total $\frac{1}{2}$
  - Notice pattern: P(landing on n) $= \frac{1}{2}$ for all $n \geq 1$
- **Symmetry Observation**:
  - Probability of landing on vs passing over should be related
  - Each successful path has total probability $\frac{1}{2^{10}}$
  - Count number of successful paths
- **Multiple Approaches**:
  - Method 1: Symmetry/intuitive argument
  - Method 2: Recursive formula
  - Method 3: Counting with stars and bars
  - Method 4: Induction proof
  - Method 5: Generating functions

C. Argument Construction

- **Logical Structure (Recursive Method)**:
  1. Define $P(n)$ = probability of landing on n
  2. Base case: $P(1) = \frac{1}{2}$
  3. Recursive formula: $P(n) = \sum_{k=0}^{n-1} P(k) \cdot \frac{1}{2^{n-k}}$
  4. Show $P(n) = \frac{1}{2}P(n-1)$ by clever manipulation
  5. Conclude $P(n) = P(n-1) = \cdots = P(1) = \frac{1}{2}$
- **Key Deductions**:
  - Sum $\sum_{m=1}^{\infty} \frac{1}{2^m} = 1$ (valid probability distribution)
  - Each path to 10 has form: visit subset of $\{1, 2, \ldots, 9\}$, then jump to 10
  - Number of subsets $= 2^9 = 512$
  - Each path has probability $\frac{1}{2^{10}}$
  - Total: $\frac{512}{1024} = \frac{1}{2}$

D. Cross-Domain Translation

- **Combinatorial**: Count compositions of 10 into positive integers
- **Algebraic**: Recursive sequence with elegant solution
- **Probabilistic**: Conditional probability and independence
- **Analytical**: Generating function approach

Tactical Selection

3. Tactical Methodology Selection

A. Core Mathematical Tactics

- **Pattern Recognition**: Calculate first few cases
- **Recursion**: Set up recursive formula for P(n)
- **Counting**: Use stars and bars or binary representation
- **Induction**: Prove P(n) = 1/2 for all $n \geq 1$
- **Symmetry**: Use probability distribution properties

B. Domain-Specific Tactics

- **Probability**:
  - Law of total probability

- Independence of jumps
- Conditioning on previous position

- **Combinatorics**:
  - Stars and bars for compositions
  - Binary representation (visit or skip positions 1-9)
  - Binomial coefficient sum: $\sum_{k=0}^{n} \binom{n}{k} = 2^n$

- **Series**:
  - Geometric series: $\sum_{m=1}^{\infty} \frac{1}{2^m} = 1$
  - Telescoping or recursive simplification

C. Tactical Priorities

1. Recognize pattern by computing P(1), P(2) (90 seconds)
2. Conjecture P(n) = 1/2 for all n (30 seconds)
3. Verify using counting method (90 seconds)
4. Select answer (E) (30 seconds)

Conversion Techniques

4. Conversion Techniques Application

A. Method 1: Counting Paths (Most Intuitive)

- **Key observation**: To reach 10, Flora must visit some subset of $\{1, 2, \ldots, 9\}$, then jump to 10
- For each position $k \in \{1, 2, \ldots, 9\}$, Flora either:
  - Lands on k (visits it)
  - Jumps over k (doesn't visit it)
- This gives $2^9 = 512$ possible paths
- **Example paths**:
  - $0 \to 10$ (one jump of 10)
  - $0 \to 1 \to 10$ (jumps of 1, then 9)
  - $0 \to 5 \to 10$ (jumps of 5, then 5)
  - $0 \to 1 \to 2 \to 3 \to 4 \to 5 \to 6 \to 7 \to 8 \to 9 \to 10$
- **Key insight**: Each path that lands on 10 has exactly the same total probability

- If path visits positions $k_1, k_2, \ldots, k_r$ before 10, the jumps are:

$$0 \to k_1, \quad k_1 \to k_2, \quad \ldots, \quad k_r \to 10$$

  with sizes $k_1, k_2 - k_1, \ldots, 10 - k_r$ and total $k_1 + (k_2 - k_1) + \cdots + (10 - k_r) = 10$
- Probability of this path: $\frac{1}{2^{k_1}} \cdot \frac{1}{2^{k_2 - k_1}} \cdots \frac{1}{2^{10 - k_r}} = \frac{1}{2^{10}}$
- Every path has probability $\frac{1}{2^{10}} = \frac{1}{1024}$
- Total probability: $\frac{512}{1024} = \frac{1}{2}$

B. Method 2: Recursive Formula

- Let $P(n)$ = probability of landing on position n
- Base case: $P(0) = 1$ (start at 0), $P(1) = \frac{1}{2}$
- Recursive formula: Can reach n from any $k < n$ with one jump

$$P(n) = \sum_{k=0}^{n-1} P(k) \cdot \frac{1}{2^{n-k}}$$

- For $n = 9$:

$$P(9) = \sum_{k=0}^{8} P(k) \cdot \frac{1}{2^{9-k}}$$

- Divide by 2:

$$\frac{P(9)}{2} = \sum_{k=0}^{8} P(k) \cdot \frac{1}{2^{10-k}}$$

- For $n = 10$:

$$P(10) = \sum_{k=0}^{9} P(k) \cdot \frac{1}{2^{10-k}} = \frac{P(9)}{2} + P(9) \cdot \frac{1}{2} = P(9)$$

- By same reasoning: $P(10) = P(9) = P(8) = \cdots = P(2) = P(1) = \frac{1}{2}$

C. Method 3: Stars and Bars

- To reach 10, need jumps that sum to 10: $j_1 + j_2 + \cdots + j_k = 10$ (all $j_i > 0$)
- Number of compositions of 10 into positive integers:

$$\sum_{k=1}^{10} \binom{10-1}{k-1} = \sum_{k=1}^{10} \binom{9}{k-1} = \sum_{j=0}^{9} \binom{9}{j} = 2^9 = 512$$

- Each composition has probability $\frac{1}{2^{10}}$ (since jumps sum to 10)
- Total probability: $\frac{512}{1024} = \frac{1}{2}$

D. Method 4: Induction

- **Claim**: $P(n) = \frac{1}{2}$ for all $n \geq 1$
- **Base case**: $P(1) = \frac{1}{2}$ (only way is jump 1 from 0)
- **Inductive step**: Assume $P(k) = \frac{1}{2}$ for all $1 \leq k < n$
- Then:

$$\begin{aligned}
P(n) &= \sum_{k=0}^{n-1} P(k) \cdot \frac{1}{2^{n-k}} \\
&= P(0) \cdot \frac{1}{2^n} + \sum_{k=1}^{n-1} \frac{1}{2} \cdot \frac{1}{2^{n-k}} \\
&= \frac{1}{2^n} + \frac{1}{2} \sum_{k=1}^{n-1} \frac{1}{2^{n-k}} \\
&= \frac{1}{2^n} + \frac{1}{2} \sum_{j=1}^{n-1} \frac{1}{2^j} \quad (\text{let } j = n-k) \\
&= \frac{1}{2^n} + \frac{1}{2} \cdot \frac{\frac{1}{2}(1 - \frac{1}{2^{n-1}})}{1 - \frac{1}{2}} \\
&= \frac{1}{2^n} + \frac{1}{2} \cdot (1 - \frac{1}{2^{n-1}}) \\
&= \frac{1}{2^n} + \frac{1}{2} - \frac{1}{2^n} = \frac{1}{2}
\end{aligned}$$

E. Method 5: Symmetry Argument

- From position 0, Flora either:
  - Lands on 10 eventually
  - Jumps past 10 without landing on it
- By symmetry, these two events are equally likely!
- **Why?** If Flora lands before 10, she faces the same problem scaled down
- The structure is self-similar, suggesting equal probabilities
- Therefore: P(land on 10) = P(jump past 10) = $\frac{1}{2}$

Verification Strategy

5. Verification Strategy Design

A. Verification Level

- **Selected Level**: Level 3 - Pattern Verification

- **Reasoning**: Check first few cases match pattern
- **Time Budget**: 60 seconds

B. Verification Checklist

- **Check P(1)**:
  - Only way: $0 \to 1$ (jump 1)
  - Probability: $\frac{1}{2}$ ✓
- **Check P(2)**:
  - Way 1: $0 \to 2$ (jump 2), probability $\frac{1}{4}$
  - Way 2: $0 \to 1 \to 2$ (jumps 1,1), probability $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$
  - Total: $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ ✓
- **Check P(3)**:
  - $0 \to 3$: $\frac{1}{8}$
  - $0 \to 1 \to 3$: $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$
  - $0 \to 2 \to 3$: $\frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$
  - $0 \to 1 \to 2 \to 3$: $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$
  - Total: $\frac{4}{8} = \frac{1}{2}$ ✓
- **Check counting**:
  - For $n = 10$: $2^9 = 512$ paths
  - Each has probability $\frac{1}{1024}$
  - Total: $\frac{512}{1024} = \frac{1}{2}$ ✓

C. Alternative Verification Methods

- **Method 1**: Verify recursive formula for several values
- **Method 2**: Check that probability distribution is valid (sums to 1)
- **Method 3**: Verify binomial sum: $\sum_{k=0}^{9} \binom{9}{k} = 2^9$

D. Answer Choice Validation

- Our answer: $\frac{1}{2}$
- Matches choice: **(E)**
- Why not the others?
  - (A), (B), (C), (D): All less than $\frac{1}{2}$, don't match the pattern
  - Pattern shows P(n) $= \frac{1}{2}$ is universal for $n \geq 1$

Solution Roadmap

6. Solution Roadmap

A. High-Level Solution Path

1. **Compute first cases**: $\mathrm{P}(1) = \frac{1}{2}$, $\mathrm{P}(2) = \frac{1}{2}$
2. **Recognize pattern**: $\mathrm{P(n)} = \frac{1}{2}$ for all $n \geq 1$
3. **Verify with counting**: 2^9 paths, each with probability $\frac{1}{2^{10}}$
4. **Calculate**: $\frac{2^9}{2^{10}} = \frac{1}{2}$
5. **Select answer**: (E)

B. Detailed Solution Steps

Step 1: Calculate probability of landing on 1.

- Only way: jump 1 from position 0
- Probability: $\frac{1}{2^1} = \frac{1}{2}$

Step 2: Calculate probability of landing on 2.

- Direct: $0 \to 2$ (jump 2), probability $\frac{1}{4}$
- Via 1: $0 \to 1 \to 2$, probability $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$
- Total: $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

Step 3: Recognize the pattern.

- $\mathrm{P}(1) = \mathrm{P}(2) = \frac{1}{2}$
- Conjecture: $\mathrm{P(n)} = \frac{1}{2}$ for all $n \geq 1$

Step 4: Verify with counting argument.

- To reach 10, must choose which of $\{1, 2, \ldots, 9\}$ to visit
- Number of choices: $2^9 = 512$
- Each path has probability $\frac{1}{2^{10}} = \frac{1}{1024}$
- (Because jump sizes sum to 10)
- Total: $\frac{512}{1024} = \frac{1}{2}$

Step 5: Select the answer.

- Answer is $\frac{1}{2}$, which is choice **(E)**

C. Key Decision Points

- **Decision 1**: Compute first few cases to find pattern
- **Decision 2**: Use counting interpretation (binary choices)
- **Decision 3**: Recognize all paths have same probability

D. Time Management

- Understanding problem: 30 seconds
- Computing P(1), P(2): 60 seconds
- Recognizing pattern: 30 seconds
- Counting argument: 90 seconds
- Verification: 30 seconds
- **Total**: 4 minutes

Competition Strategy

7. Competition Strategy Integration

A. Problem Triage

- **Difficulty Assessment**: Upper-medium (Problem 17/25)
- **Confidence Level**: High if you spot the pattern quickly
- **Decision**: Worth attempting - elegant answer suggests quick solution

B. Time Allocation

- **Target Time**: 3-4 minutes
- **Maximum Time**: 5 minutes
- **Quick Strategy**: If you see $P(1) = P(2) = \frac{1}{2}$, guess (E)

C. Solution Format

- Multiple choice: Select **(E)** $\frac{1}{2}$
- Verification: Check first 2-3 cases match pattern
- Bubble answer sheet carefully

D. Strategic Considerations

- **Pattern Recognition**: Computing P(1) and P(2) reveals the answer

- **Answer Choice Clue**: (E) $\frac{1}{2}$ is suspiciously simple
- **Elimination**: Can't easily eliminate without computation
- **Time Saver**: Don't need rigorous proof in competition

Common Pitfalls and Warnings

Common Pitfalls and Warnings

A. Conceptual Errors

- **Counting only one path**: Forgetting multiple ways to reach 10
- **Wrong probability model**: Not using $\frac{1}{2^m}$ correctly
- **Stopping at finite sum**: Not considering all possible paths
- **Confusing land on vs pass through**: Must land exactly on 10

B. Computational Errors

- **Arithmetic**: $\mathrm{P}(2) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$, not $\frac{1}{4}$
- **Binomial sum**: $\sum_{k=0}^{9} \binom{9}{k} = 2^9 = 512$, not 1024
- **Simplification**: $\frac{512}{1024} = \frac{1}{2}$, not other fractions

C. Strategic Errors

- **Overcomplicating**: Don't need full recursion or generating functions
- **Missing pattern**: Not computing enough cases to see $\mathrm{P(n)} = \frac{1}{2}$
- **Time management**: Should take 3-5 minutes max

D. Warning Signs

- If you get answer other than $\frac{1}{2}$, recheck pattern
- If calculation is too complex, look for simpler approach
- The elegant answer (E) is a strong hint

Detailed Solution

A. Main Solution (Counting Method)

1. **Observation**: To reach 10, Flora chooses which positions in $\{1, 2, \ldots, 9\}$ to land on
2. **Binary choice**: For each position $k \in \{1, 2, \ldots, 9\}$:
    - Either land on k (visit it)

- Or jump over k (don't visit it)

3. **Count paths**: $2^9 = 512$ possible paths

4. **Key insight**: Each path has the same probability!

   - If path visits positions $0 < k_1 < k_2 < \cdots < k_r < 10$
   - Jump sizes: $k_1, k_2 - k_1, \ldots, k_r - k_{r-1}, 10 - k_r$
   - Sum of jumps: $k_1 + (k_2 - k_1) + \cdots + (10 - k_r) = 10$
   - Probability: $\frac{1}{2^{k_1}} \cdot \frac{1}{2^{k_2 - k_1}} \cdots \frac{1}{2^{10-k_r}} = \frac{1}{2^{10}}$

5. **Total probability**:

$$P(\text{land on } 10) = 512 \times \frac{1}{2^{10}} = \frac{512}{1024} = \frac{1}{2}$$

B. Alternative Solution (Pattern Recognition)

1. Calculate P(1): Only way is $0 \to 1$

$$P(1) = \frac{1}{2}$$

2. Calculate P(2):

   - $0 \to 2$: probability $\frac{1}{4}$
   - $0 \to 1 \to 2$: probability $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$
   - Total: $P(2) = \frac{1}{2}$

3. Calculate P(3):

   - $0 \to 3$: $\frac{1}{8}$
   - $0 \to 1 \to 3$: $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$
   - $0 \to 2 \to 3$: $\frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$
   - $0 \to 1 \to 2 \to 3$: $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$
   - Total: $P(3) = \frac{1}{2}$

4. Recognize pattern: $P(n) = \frac{1}{2}$ for all $n \geq 1$

5. Therefore: $P(10) = \frac{1}{2}$

C. Verification

- Pattern: P(1) = P(2) = P(3) = $\frac{1}{2}$ ✓
- Counting: $2^9 = 512$ paths, each probability $\frac{1}{1024}$ ✓
- Total: $\frac{512}{1024} = \frac{1}{2}$ ✓

Key Insights

Key Insight: This problem has a beautiful result: the probability of landing on ANY positive integer n is always $\frac{1}{2}$!
Why this works:

- To reach position n, Flora must choose which of positions $\{1, 2, \ldots, n-1\}$ to visit
- This is a binary choice for each position: visit or skip
- Number of paths: 2^{n-1}
- Each path to n has jump sizes that sum to n
- Therefore, each path has probability $\frac{1}{2^n}$
- Total probability: $\frac{2^{n-1}}{2^n} = \frac{1}{2}$

General Pattern: For probability distribution P(jump m) $= \frac{1}{2^m}$:

- Distribution is valid: $\sum_{m=1}^{\infty} \frac{1}{2^m} = 1$
- Probability of landing on any $n \geq 1$ is $\frac{1}{2}$
- Probability of jumping past n (never landing on it) is also $\frac{1}{2}$
- This creates a beautiful symmetry

Recursive interpretation:

$$P(n) = \sum_{k=0}^{n-1} P(k) \cdot \frac{1}{2^{n-k}}$$

can be rearranged to show $P(n) = P(n-1)$ for $n \geq 2$, confirming the constant value.
Why answer (E) is elegant: The simple answer $\frac{1}{2}$ reflects the deep symmetry in the problem structure. From any position, the probability of landing on the next target equals the probability of skipping it entirely.

Final Answer

The probability that Flora will eventually land at 10 is:

$$\boxed{\textbf{(E)}\ \frac{1}{2}}$$

Solution Summary:

- Each path to 10 visits some subset of $\{1, 2, \ldots, 9\}$
- Number of subsets: $2^9 = 512$
- Each path has probability $\frac{1}{2^{10}} = \frac{1}{1024}$

- Total probability: $\frac{512}{1024} = \frac{1}{2}$

Confidence Assessment

- **Confidence Level**: 99%
  - Pattern confirmed for P(1), P(2), P(3)
  - Counting argument is rigorous
  - Beautiful mathematical result
  - Multiple solution methods all agree
- **Verification Applied**: Level 3 - Pattern Verification
  - Verified first three cases
  - Confirmed counting: $2^9 = 512$ paths
  - Checked simplification: $\frac{512}{1024} = \frac{1}{2}$
- **Time-Confidence Trade-off**: High confidence in 3-4 minutes